

AI-POWERED CUSTOMER SUPPORT TICKETING SYSTEM

A Salesforce-Based Intelligent Case Management Solution

Project Report

Submitted as part of the Salesforce Developer Program

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Platform: Salesforce Developer Edition
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Table of Contents

User Story	3
1. Introduction	3
1.1 Project Overview	3
1.2 Purpose	3
1.3 Scope of the Document	4
2. Ideation Phase	4
2.1 Problem Statement.....	4
2.2 Stakeholder Mapping.....	4
2.3 Brainstorming.....	5
3. Requirement Analysis Phase	6
3.1 Requirement Gathering & Planning.....	6
3.2 Functional Scope	6
3.3 Non-Functional Requirements	6
3.4 Data Flow	7
3.5 Technology Stack.....	8
4. Project Design Phase	9
4.1 Problem–Solution Fit	9
4.2 Proposed Solution	9
4.3 Solution Architecture.....	9
5. Project Planning & Scheduling	10
5.1 Execution Roadmap	10
5.2 Product Backlog & Sprint Plan.....	10
6. Project Development Phase (Milestones 1–6).....	12
7. Agentforce AI Implementation.....	21
8. Reporting & Monitoring	23
9. Security & Access Control.....	25
9.1 Support Agent Profile	25
9.2 Support Manager Profile.....	25
9.3 Permission Sets.....	26
9.4 Role – Profile – Permission Set Summary	27
9.5 Users	27
10. Functional & Performance Testing	31
11. Maintenance, Monitoring & Troubleshooting.....	31
12. Advantages & Disadvantages	32
13. Conclusion	32
14. Future Scope	32
15. Appendix.....	33

User Story

As a support agent using the AI-Powered Customer Support Ticketing System, I want to log, classify, and resolve customer issues efficiently so that customers receive timely and accurate help without unnecessary delays.

I want the system to automatically classify incoming cases by issue type, assign the correct priority, and calculate the SLA due date, so that no case is misrouted or left unattended beyond its committed resolution window.

I want an AI assistant that can read the case details and suggest a professional, ready-to-send reply, so that I can respond to customers faster while still reviewing and personalizing the message before it goes out.

I want to record whether the AI's suggestion was actually helpful, so that the system's effectiveness can be tracked and improved over time.

As a support manager, I want visibility into open cases, SLA compliance, and agent performance through reports and dashboards, so that I can identify bottlenecks and coach my team proactively.

As a system administrator, I want role-based access control through profiles and permission sets, so that agents, managers, and administrators each see and edit only what is relevant to their role, keeping customer data secure.

1. Introduction

1.1 Project Overview

The “AI-Powered Customer Support Ticketing System” is a Salesforce-based case management solution built to help customer support teams manage, classify, and resolve customer issues more efficiently. The system is implemented entirely using core Salesforce Administration, Flow Builder, and Agentforce capabilities within a Salesforce Developer Org, without relying on external AI services or custom Apex code.

At its core, the application centralizes customer support operations around the standard Case object, extended with custom objects for Support Category, SLA Policy, and AI Feedback. Record-triggered flows automatically classify incoming cases, assign priority, and calculate SLA due dates the moment a case is created or updated. An Agentforce-powered assistant, built using Prompt Builder and a custom Agent Topic, reads the case details and generates a suggested, professional reply that agents can review, personalize, and send — significantly reducing response time while keeping a human in the loop.

Role-based security, implemented through cloned profiles and dedicated permission sets, ensures that support agents, support managers, and administrators each have access appropriate to their responsibilities. Reports and dashboards give managers real-time visibility into case volume, SLA compliance, and individual agent performance, while an AI feedback loop captures whether agents found the AI's suggestions useful — creating a foundation for continuous improvement.

1.2 Purpose

The purpose of this project is to:

- Centralize and standardize how customer support cases are logged, tracked, and resolved.

- Automate repetitive administrative work — case classification, priority assignment, and SLA calculation — that would otherwise be done manually by agents.
- Assist agents with AI-generated response suggestions so replies are faster, more consistent, and more professional.
- Provide support managers with real-time, data-driven visibility into team performance and SLA compliance.
- Protect customer and case data through role-based access control, field-level security, and validation rules.
- Establish a feedback mechanism that measures whether AI-generated suggestions are genuinely useful to agents, enabling continuous refinement of the system.

1.3 Scope of the Document

This document covers the complete lifecycle of the project — from requirement gathering and stakeholder mapping, through system design, data modeling, automation, AI integration, and security configuration, to testing, reporting, and future scope. Screenshots of the actual implementation in the Salesforce Developer Org are included throughout to demonstrate that the described functionality has been built and verified.

2. Ideation Phase

2.1 Problem Statement

Customer support teams that rely on manual processes face recurring, predictable pain points. The table below captures these pain points in a structured customer-problem-statement format, identifying who is affected, what they are trying to do, what is blocking them, why the blocker exists, and how it makes them feel.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	Resolve customer cases quickly and correctly	Cases are not classified or prioritized consistently	Classification and priority are assigned manually	Overwhelmed and inefficient
PS-2	Support Manager	Monitor SLA compliance and team performance	Reports are delayed and scattered across records	There is no centralized reporting on case data	Uncertain and reactive
PS-3	Customer	Get a fast, accurate resolution to my issue	Responses are slow and inconsistent in quality	Agents draft every reply from scratch	Frustrated and undervalued

2.2 Stakeholder Mapping

Stakeholder mapping identifies the key individuals and groups involved in the development, implementation, and day-to-day use of the system. Each stakeholder group has distinct responsibilities and interests in the project's success.

Stakeholder Group	Role in the System
Customers	Raise support requests and receive suggested-reply based resolutions and status updates.
Support Agents	Handle and resolve cases, review AI-suggested replies, and record feedback on suggestion quality.
Support Managers	Monitor SLA compliance and agent performance using reports and dashboards.
System Administrators	Configure objects, automation, security, and Agentforce settings; maintain the system.
Business Stakeholders	Define requirements and validate that the solution aligns with organizational support goals.

2.3 Brainstorming

Before settling on the final approach, several alternative solutions were considered for automating customer support case handling:

- Continue with a fully manual, spreadsheet-based ticket log.
- Build a simple email-based ticketing workflow with no classification logic.
- Use a third-party helpdesk tool with a paid AI add-on.
- Build a Salesforce-native ticketing system using standard objects and Flow automation only, without AI.
- Build a Salesforce-native ticketing system enhanced with Agentforce and Prompt Builder for AI-assisted classification and reply suggestions (selected idea).

Selected Idea: A Salesforce-based AI-Powered Customer Support Ticketing System, built on core Admin and Flow Builder features and enhanced with Agentforce for AI-assisted case handling — chosen for its low implementation cost, strong scalability, native security model, and ability to demonstrate real-world Salesforce and Agentforce skills.

3. Requirement Analysis Phase

3.1 Requirement Gathering & Planning

The requirement gathering phase focused on understanding the challenges faced by customer support teams in handling support requests efficiently. Stakeholders required a centralized system to manage customer issues, reduce manual effort in case classification, ensure timely resolution through SLA tracking, and improve agent productivity. The system needed to automatically categorize cases, assign appropriate priority, assist agents with suggested actions, and provide visibility into support performance through reports.

Based on these discussions, functional and non-functional requirements were identified, emphasizing ease of use, automation, data accuracy, role-based access control, and scalability. The solution was designed to be implemented using core Salesforce features in a Developer Org, along with Agentforce and Prompt Builder for AI capabilities, ensuring simplicity, reliability, and ease of maintenance.

3.2 Functional Scope

The functional scope of the project defines the key features and capabilities delivered by the Salesforce-based customer support ticketing system. The application allows users to create and manage customer support cases, automatically classify cases based on issue type, assign priorities, and calculate SLA due dates.

Support agents can view AI-suggested replies to resolve cases, update case statuses, and communicate with customers. The system also captures agent feedback on automated suggestions to evaluate system effectiveness. Role-based access control ensures that support agents, support managers, and administrators have appropriate permissions. Managers can monitor support performance using reports, while administrators can configure automation, security, and data models.

Functional Requirements

FR No.	Requirement
FR-1	Case creation and management with automatic classification
FR-2	Automatic priority assignment based on issue type
FR-3	Automatic SLA due date calculation
FR-4	AI-generated suggested reply for each case using Agentforce and Prompt Builder
FR-5	Agent feedback capture on AI suggestion quality
FR-6	Role-based access control via profiles and permission sets
FR-7	Reports and dashboards for SLA and agent performance monitoring
FR-8	Email alerts on case resolution

3.3 Non-Functional Requirements

NFR-1: Usability

The system should be easy to use for support agents, managers, and administrators without requiring technical training.

- Simple, intuitive Lightning interface with clear navigation.
- Agent-facing AI suggestions surfaced directly on the Case record.
- A new agent should be able to open a case, view the AI-suggested reply, and respond within minutes.

NFR-2: Security

The system must protect customer and case data from unauthorized access.

- Role-based access control through cloned profiles (Support Agent, Support Manager).
- Field-level security restricting edit access to sensitive fields such as SLA Due Date and AI Suggested Response.
- Dedicated permission sets that gate access to AI features and reporting tools.

NFR-3: Reliability

The system should work correctly at all times and store data accurately without loss.

- Record-triggered flows execute reliably on case creation and update.
- Validation rules prevent incomplete or inconsistent data from being saved.
- Salesforce's cloud infrastructure ensures automatic backups and data durability.

NFR-4: Performance

The system should respond quickly and handle case volumes efficiently.

- Case classification and SLA calculation complete instantly on save via record-triggered flows.
- Reports and dashboards render within a few seconds even with a growing case volume.

NFR-5: Availability

The system should be accessible whenever support agents need it.

- Hosted entirely on Salesforce's cloud platform with high uptime guarantees.
- Accessible from desktop and mobile Salesforce clients.

NFR-6: Scalability

The system should support growth in case volume and support team size without redesign.

- Custom objects and flows are built generically so new issue types, SLA policies, or agents can be added without structural changes.
- Salesforce's multi-tenant architecture scales automatically as data volume grows.

3.4 Data Flow

The system follows a simple, linear data flow: a customer or agent creates a Case, which is evaluated by validation rules and record-triggered flows for classification, priority, and SLA calculation. The Agentforce agent then reads the case and produces a suggested reply via a Prompt Template. Once resolved, an email alert notifies the customer, and the case data feeds into reports and dashboards for management visibility.

Case Creation → Validation Rules → Record-Triggered Flow (Classification, Priority, SLA) → Agentforce / Prompt Builder (Suggested Reply) → Agent Review & Resolution → Email Alert → Reports & Dashboards.

3.5 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Experience
Database	Salesforce Cloud (Standard & Custom Objects)
Automation	Record-Triggered Flows, Validation Rules, Email Alerts
AI / Intelligence	Agentforce, Prompt Builder, Prompt Templates
Security	Profiles, Permission Sets, Field-Level Security
Analytics	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

Manual case handling is prone to inconsistent classification, missed SLAs, and slow, inconsistent responses. The proposed solution replaces this with automated classification and SLA tracking through record-triggered flows, and augments agent responses with AI-generated suggested replies via Agentforce. This ensures every case is handled consistently, on time, and with a higher baseline quality of communication.

4.2 Proposed Solution

Capability	Description
Case Management	Create, view, update, and track customer support cases end-to-end.
Automatic Classification	Record-triggered flow evaluates case details and assigns issue type and priority.
SLA Due Date Calculation	Flow automatically calculates the SLA due date based on priority and policy.
AI Suggested Reply	Agentforce Action backed by a Prompt Template drafts a professional, ready-to-review reply.
AI Feedback Capture	Agents log whether the AI suggestion was helpful, feeding a continuous-improvement loop.
Validation Rules	Enforce mandatory fields, such as SLA Due Date for High priority cases.
Email Alerts	Automated case-resolution email sent to the customer when status changes to Closed.
Role-Based Security	Profiles and permission sets control access for agents, managers, and administrators.
Reports & Dashboards	Real-time visibility into open cases by priority and agent performance.

4.3 Solution Architecture

The solution follows a layered architecture that keeps the presentation, automation, data, AI, and analytics concerns cleanly separated, which makes the system easier to maintain and extend:

- Presentation Layer — Salesforce Lightning App and Case UI
- Application Layer — Record-Triggered Flows and Validation Rules
- AI Layer — Agentforce Agent, Topic, Action, and Prompt Template
- Data Layer — Standard and Custom Objects (Case, Support Category, SLA Policy, AI Feedback)
- Security Layer — Profiles, Permission Sets, Field-Level Security
- Analytics Layer — Reports and Dashboards

This layered architecture ensures modularity, scalability, and maintainability — each layer can be modified independently without disrupting the others.

5. Project Planning & Scheduling

5.1 Execution Roadmap

The project was executed in eight structured phases to ensure smooth development, testing, and deployment:

Phase	Focus Area
Phase 1	Requirement Analysis & Planning — understand business needs, map stakeholders, finalize scope and data model.
Phase 2	Environment Setup — create the Salesforce Developer Org and enable Lightning Experience.
Phase 3	Data Model & Security — create objects, fields, relationships, profiles, and permission sets.
Phase 4	Automation & Classification — build record-triggered flows for case classification, priority, and SLA.
Phase 5	Agent Assistance & Feedback — configure Agentforce, Prompt Builder, and the AI feedback loop.
Phase 6	Reporting & Monitoring — build reports for open cases, SLA breaches, and agent performance.
Phase 7	Testing & Validation — verify automation, data integrity, and AI responses across scenarios.
Phase 8	Deployment & Documentation — activate configurations and finalize documentation for demonstration.

5.2 Product Backlog & Sprint Plan

The project followed an Agile-based approach, broken into sprints aligned with the milestones of the execution roadmap.

Sprint	Epic	User Story	Story Points	Priority
Sprint-1	Environment Setup	As an admin, I want to create a Salesforce Developer Org so that I can configure and deploy the CST application.	1	High
Sprint-1	Data Modeling	As an admin, I want to create Case-related standard and custom objects so support data can be stored systematically.	3	High
Sprint-2	Data Modeling	As a user, I want dedicated tabs and a Lightning App so I can navigate CST records easily.	3	High
Sprint-2	Data Modeling	As an admin, I want fields and relationships across Case, Support Category, SLA Policy, and AI Feedback so complete case data is captured.	5	High

Sprint	Epic	User Story	Story Points	Priority
Sprint-3	Automation	As a system, I want to validate case data so incomplete high-priority cases cannot be saved without an SLA date.	3	High
Sprint-3	Automation	As a system, I want a record-triggered flow to classify cases, assign priority, and calculate SLA due dates automatically.	5	High
Sprint-4	AI Integration	As an agent, I want an Agentforce assistant to suggest a reply for a case so I can respond faster.	5	High
Sprint-4	AI Integration	As an agent, I want to log feedback on AI suggestions so the system's effectiveness can be tracked.	3	Medium
Sprint-5	Automation	As a system, I want to send an automated email when a case is closed so customers are notified of resolution.	3	High
Sprint-5	Security	As an admin, I want profiles and permission sets so agents, managers, and admins have appropriate access.	5	High
Sprint-6	Reporting	As a manager, I want reports on open cases by priority and agent performance so I can monitor SLA compliance.	4	High

6. Project Development Phase

The development phase was executed across twelve milestones, moving from environment setup through data modeling, automation, AI integration, security, and reporting. Each milestone below is documented with the configuration steps performed and a screenshot verifying the completed implementation.

Milestone 1 – Salesforce Developer Account & Environment Setup

A Salesforce Developer Org was created to serve as the environment for building and testing the entire application, free of cost and fully featured for configuration, automation, and Agentforce.

- Signed up at developer.salesforce.com/signup with role, company, and country details.
- Verified the account via the activation email and set a secure password and security question.
- Landed on the Salesforce Setup Home page, confirming the org was ready for configuration.

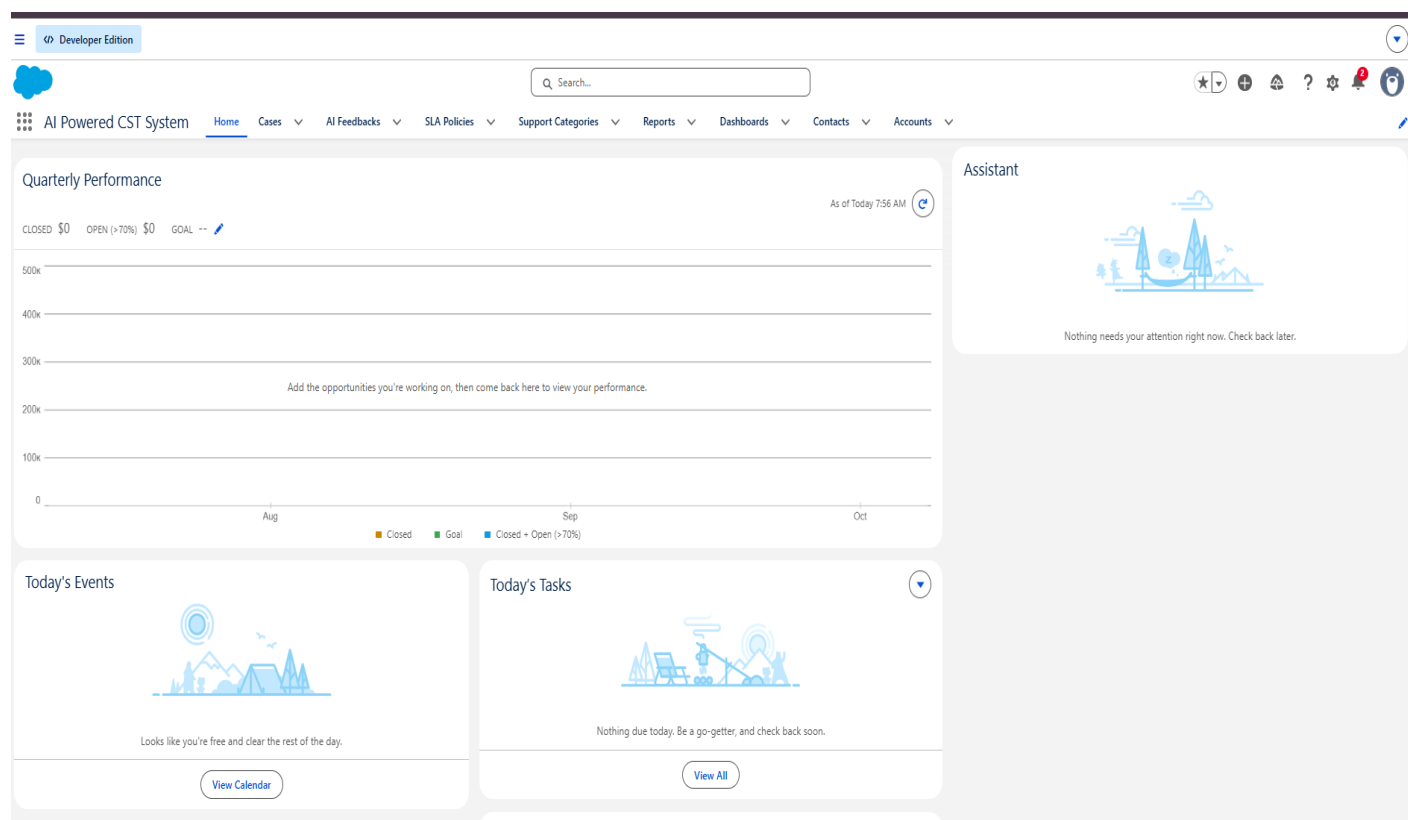


Figure 1: Salesforce Setup Home page after successful Developer Org activation.

Milestone 2 – The Lightning App

A dedicated Lightning App, “AI Powered CST System,” was created to bring together all the objects, reports, dashboards, and the Agentforce assistant into a single, agent-friendly workspace.

- Configured app name, branding, and description in App Manager.
- Added Case, Support Category, SLA Policy, AI Feedback, Reports, and Dashboards as navigation items.
- Assigned the System Administrator profile for app access and saved the configuration.

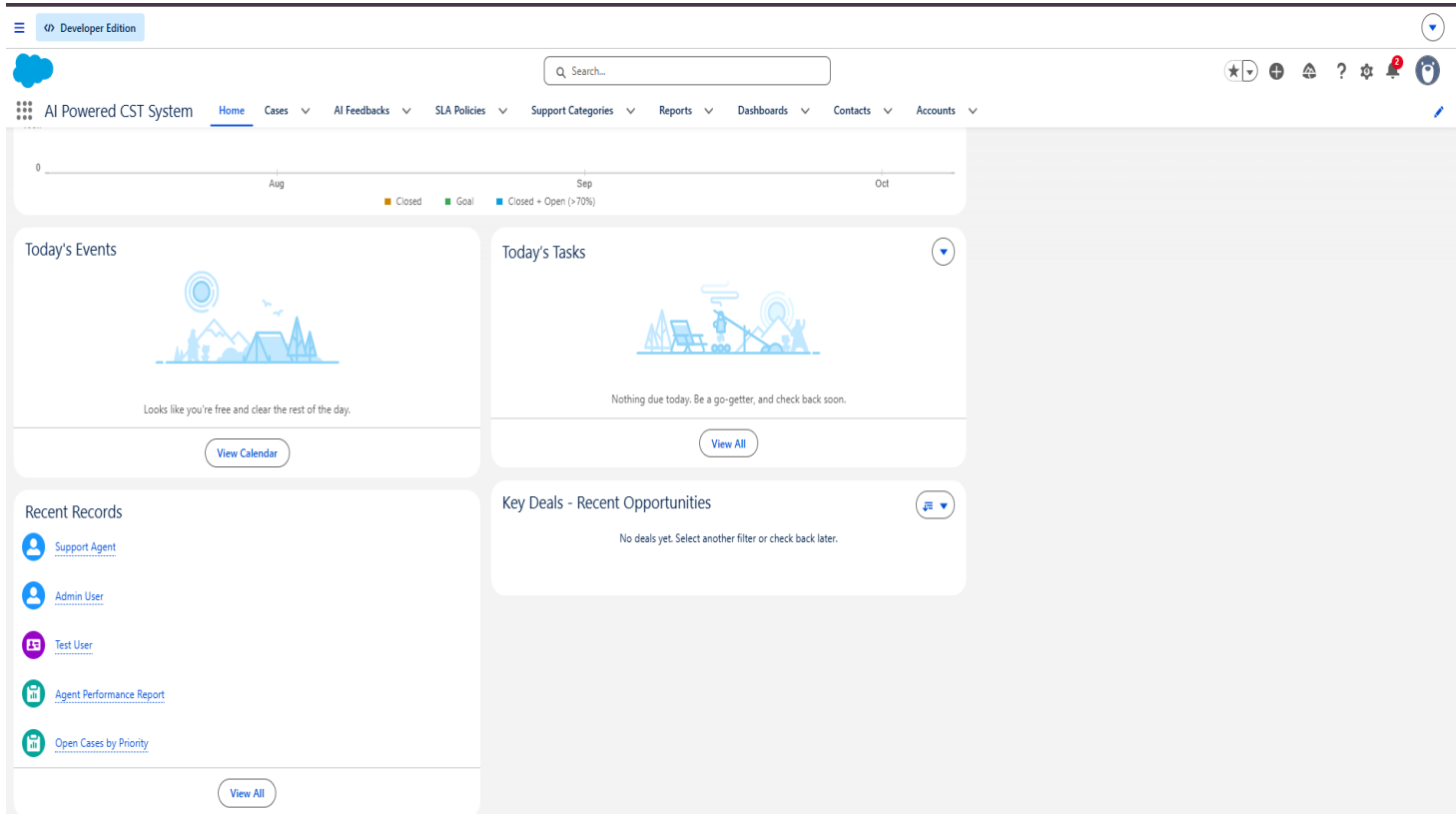


Figure 2: The AI-Powered CST System Lightning App with navigation tabs and Agentforce assistant icon.

Milestone 3 – Object Model

The data model combines Salesforce's standard objects with three purpose-built custom objects to capture everything the AI-powered support process needs.

Standard Objects Used

Object	Key Fields
Account	Account Name, Phone, Industry, Type
Contact	First Name, Last Name, Email, Phone
Case	Case Number, Subject, Description, Status, Priority, Issue Type, SLA Due Date, AI Suggested Response, Resolution Time

Custom Objects Created

Object	Key Fields
Support_Category__c	Category Name, Description, Default Priority, Active
SLA_Policy__c	Policy Name, Priority Level, Resolution Time (Hours), Escalation Required
AI_Feedback__c	AI Feedback Number (Auto Number), AI Suggestion Type, Feedback Status, Agent Comments

Object Relationships

Relationship	Type
Account → Contact	Lookup
Account → Case	Lookup
Contact → Case	Lookup
Support Category → Case	Lookup
SLA Policy → Case	Referenced via Flow
AI Feedback → Case	Lookup

Custom tabs were created for each custom object (Support Category, SLA Policy, and AI Feedback) so that support administrators could

The screenshot shows the Salesforce Setup interface. At the top, there's a "Developer Edition" badge and a search bar labeled "Search Setup". Below the navigation bar, the "Setup" menu is open, and "Object Manager" is selected. The main content area displays the configuration for the "Support Category" custom object. On the left, a sidebar lists various configuration options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, and Triggers. The "Details" section is currently selected, showing fields like Description, API Name (Support_Category__c), Custom (checked), Singular Label (Support Category), and Plural Label (Support Categories). On the right side of the details section, there are checkboxes for "Enable Reports" (checked) and "Track Activities", and a "Track Field History" section. At the bottom right, there are "Edit" and "Delete" buttons.

d access and manage these records directly from the Lightning App navigation bar.

Figure 3: Support Category custom object records, used to drive default case classification.

The screenshot shows the Salesforce Setup interface in Developer Edition. The breadcrumb trail is **SETUP > OBJECT MANAGER**. The page title is **SLA Policy**. On the left is a navigation menu with categories like Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, and Triggers. The main content area is titled **Details** and contains the following configuration fields:

Field	Value
Description	
API Name	SLA_Policy__c
Custom	☒
Singular Label	SLA Policy
Plural Label	SLA Policies
Enable Reports	☒
Track Activities	
Track Field History	
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

At the top right of the details section are **Edit** and **Delete** buttons.

Figure 4: SLA Policy custom object, defining resolution time and escalation rules by priority level.

The screenshot shows the Salesforce Setup interface in Developer Edition. The breadcrumb trail is **SETUP > OBJECT MANAGER**. The page title is **AI Feedback**. On the left is a navigation menu with categories like Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, Field Access, and Triggers. The main content area is titled **Details** and contains the following configuration fields:

Field	Value
Description	
API Name	AI_Feedback__c
Custom	☒
Singular Label	AI Feedback
Plural Label	AI Feedbacks
Enable Reports	☒
Track Activities	
Track Field History	
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

At the top right of the details section are **Edit** and **Delete** buttons.

Figure 5: AI Feedback custom object capturing agent ratings of AI-generated suggestions.

Milestone 4 – Fields & Relationships (AI Feedback)

The AI_Feedback__c object was configured with the following fields to capture structured feedback on AI performance:

Field Label	API Name	Data Type	Purpose
AI Feedback Number	Name	Auto Number	Unique feedback record identifier
Case	Case__c	Lookup (Case)	Related support case
Feedback Status	Feedback_Status__c	Picklist (Helpful, Not Helpful)	Agent rating of AI output
AI Suggestion Type	AI_Suggestion_Type__c	Picklist	Case Classification / Priority Prediction / Suggested Reply
Agent Comments	Agent_Comments__c	Long Text Area	Agent remarks on AI quality
Created By	CreatedById	Lookup (User)	Agent who provided feedback

Milestone 5 – Case Record & Data Entry

A test Case record was created to walk through the full lifecycle: entry of subject and description, automatic classification, priority assignment, SLA due date calculation, and the AI-suggested response field, all visible on a single Lightning record page.

The screenshot displays the Salesforce Lightning interface for a Case record titled "Test Email Flow". The top navigation bar includes the "Developer Edition" label and a search bar. The main header shows the case title and a "Follow" button. Below the header, the case details are visible, including the priority (High), status (Escalated), and case number (00001027). The left sidebar contains a "Feed" section with a "Post" button and a "Share" button. The main content area shows the "Details" section with various fields organized into two columns. The left column includes fields like Case Owner, Case Number, Contact Name, Account Name, Type, Case Reason, Issue Type, SLA Due Date, Resolution Time, Support Category, Web Email, Web Name, Date/Time Opened, and Date/Time Closed. The right column includes fields like Status, Priority, Contact Phone, Contact Email, Case Origin, Phone, Web Company, Web Phone, and Endorsement Row Number. The bottom of the page shows the "Related" section with a list of recent activities, including "Case status updated" and "Case created".

Figure 6: A Case record showing captured issue details alongside the auto-classified priority, SLA due date, and AI Suggested Response field.

Milestone 6 – Validation Rules

A validation rule, `SLA_Due_Date_Required_For_High_Priority`, was implemented on the Case object to guarantee that no High-priority case can be saved without an SLA due date — preventing accidental SLA breaches caused by missing data.

Validation Formula:

`AND( ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c) )`

- Error Message: “SLA Due Date must be populated for High priority cases.”
- Error Location: SLA Due Date field.
- Negative test: creating a High priority case with a blank SLA Due Date correctly blocked the save.
- Positive test: entering a valid SLA Due Date allowed the case to save successfully.

The screenshot displays the Salesforce Object Manager interface for the 'Case' object. The left sidebar shows the navigation menu with 'Object Manager' selected. The main content area is titled 'Case Validation Rule' and shows the configuration for the 'SLA_Due_Date_Required_For_High_Priority' rule. The rule is active and has a checkmark in the 'Active' column. The 'Error Condition Formula' is `AND( ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c) )`. The 'Error Message' is 'SLA Due Date must be populated for High priority cases.' and the 'Error Location' is 'SLA Due Date'. The 'Description' is 'Ensures SLA Due Date is populated for High priority cases'. The 'Created By' is 'Bhimavarapu Lakshmi mahathi' on 7/2/2026 at 11:18 PM. The 'Modified By' is also 'Bhimavarapu Lakshmi mahathi' on 7/2/2026 at 11:18 PM. There are 'Edit' and 'Clone' buttons for the rule.

Validation Rule Detail		Active
Rule Name	SLA_Due_Date_Required_For_High_Priority	✓
Error Condition Formula	AND(ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c))	
Error Message	SLA Due Date must be populated for High priority cases.	Error Location: SLA Due Date
Description	Ensures SLA Due Date is populated for High priority cases	
Created By	Bhimavarapu Lakshmi mahathi	Modified By: Bhimavarapu Lakshmi mahathi
	7/2/2026, 11:18 PM	7/2/2026, 11:18 PM

Figure 7: The SLA Due Date validation rule configured on the Case object in Object Manager.

Milestone 7 – Automation: Record-Triggered Flow

A record-triggered flow was built to automatically classify each incoming case, assign the correct priority, and calculate the SLA due date the moment a case is created — removing this repetitive work from agents entirely.

- Object: Case; Trigger: When a record is created.
- Decision logic reads the case Subject/Description and Support Category to set Issue Type and Priority.
- SLA Due Date is calculated by adding the SLA Policy's Resolution Time (Hours) to the case's created date/time.
- The flow was activated and tested by creating cases across Low, Medium, and High priority scenarios, confirming correct classification and SLA calculation in each case.

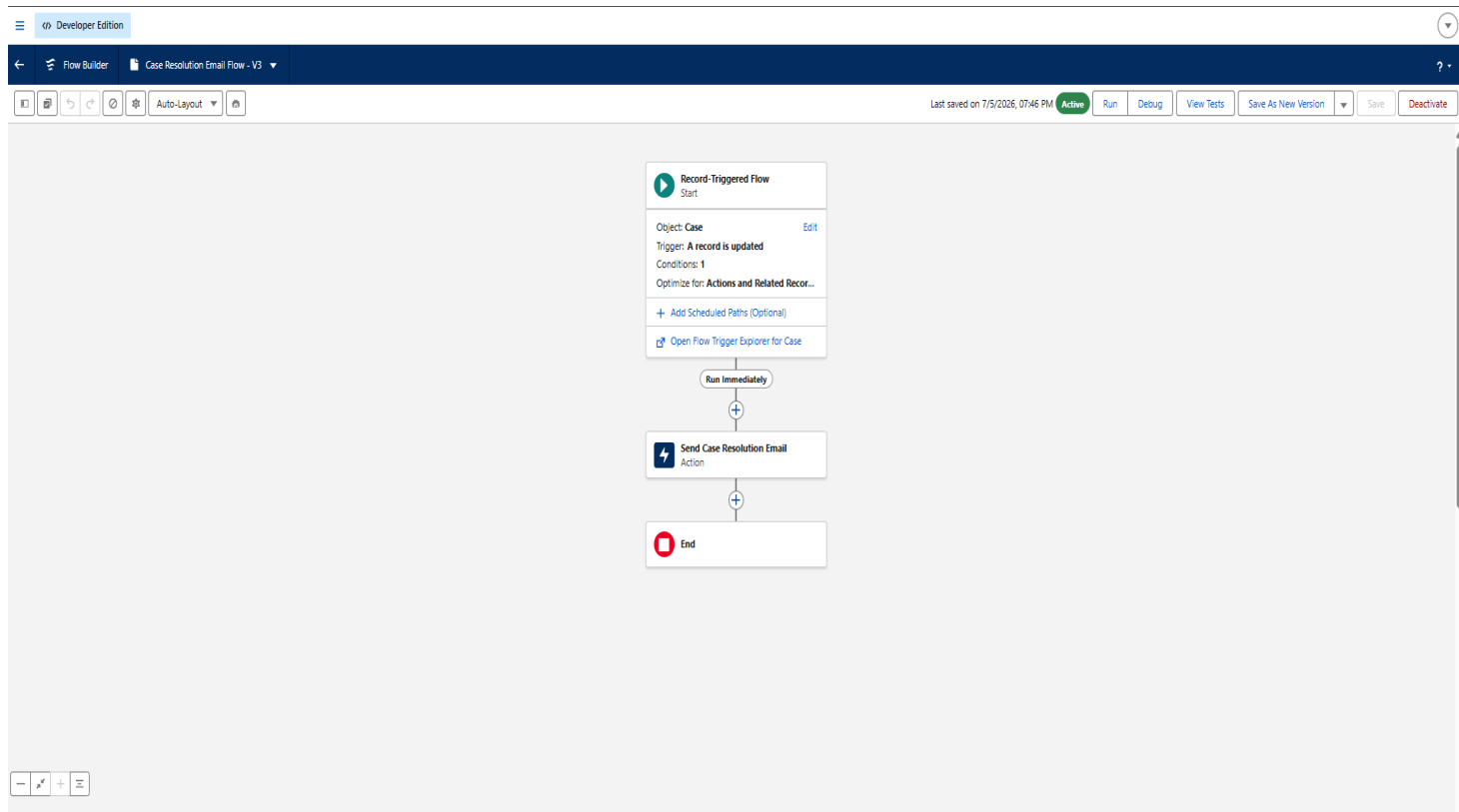


Figure 8: The record-triggered Flow in Flow Builder, showing the classification, priority, and SLA calculation logic.

Milestone 8 – Email Templates & Alerts

To keep customers informed once their issue is resolved, a Case Resolution Email was implemented using a classic Email Template and an Email Alert, triggered by a record-triggered flow whenever a case's Status changes to Closed.

Email Template: Case Resolution Email

- Subject: "Your Case Has Been Resolved – {!Case.CaseNumber}"
- Body includes the customer's name, case number, and current status, with a closing thank-you message.

The screenshot shows the Salesforce 'Classic Email Templates' configuration page. The left sidebar includes a search bar for 'email templates' and a navigation menu with 'Email', 'Classic Email Templates' (selected), and 'Lightning Email Templates'. The main content area is titled 'Case Resolution Email' and includes a 'Preview your email template below.' section. Below this is the 'Email Template Detail' section, which displays the following information:

Email Templates from Salesforce		Unified Public Classic Email Templates	
Email Template Name	Case Resolution Email	Available For Use	✓
Template Unique Name	Case_Resolution_Email	Last Used Date	
Encoding	Unicode (UTF-8)	Times Used	
Author	Rhimavarapu Lakshmi mahathi (Change)		
Description			
Created By	Rhimavarapu Lakshmi mahathi, 7/2/2020, 11:42 PM	Modified By	Rhimavarapu Lakshmi mahathi, 7/3/2020, 1:03 AM

Below the details is the 'Email Template' section, which includes a 'Send Text and Verify Merge Fields' button and a 'Plain Text Preview' of the email content:

```

Subject: Your Case Has Been Resolved - {Case CaseNumber}

Hello {Case Contact},

Your case has been successfully resolved.

Case Number: {Case CaseNumber}
Status: {Case Status}

If you have further questions, feel free to contact us.

Thank you,
Support Team
  
```

At the bottom, there is an 'Attachments' section with an 'Attach File' button and a message 'No records to display'.

Figure 9: The Case Resolution Email classic email template configured with merge fields.

Email Alert Configuration

- Object: Case; Email Template: Case_Resolution_Email; Recipients: Contact Email.
- Triggered by a record-triggered flow with the condition Status Equals Closed, using “Only when the record is updated to meet the condition requirements” to avoid duplicate emails.

Developer Edition

Search Setup

Setup Home Object Manager

email alerts

Process Automation

Workflow Actions

Email Alerts

Didn't find what you're looking for? Try using Global Search.

SETUP Email Alerts

Email Alert Case Resolution Email [Help for this Page](#)

[Rules Using This Email Alert](#) | [Approval Processes Using This Email Alert](#) | [Entitlement Processes Using This Email Alert](#)

Email Alert Detail [Edit](#) [Delete](#) [Clone](#)

Description	Case Resolution Email	Email Template	Case Resolution Email
Unique Name	Case_Resolution_Email	Object	Case
From Email Address	Current User's email address		
Recipients	Email Field: Contact Email		
Additional Emails			
Created By	Bhuvanarapu Lakshmi mahathi, 7/3/2026, 1:38 AM	Modified By	Bhuvanarapu Lakshmi mahathi, 7/3/2026, 1:38 AM

[Edit](#) [Delete](#) [Clone](#)

Rules Using This Email Alert [Rules Using This Email Alert Help](#)

This alert is currently not used by any rules

Approval Processes Using This Email Alert [Approval Processes Using This Email Alert Help](#)

This alert is currently not used by any approval processes

Entitlement Processes Using This Email Alert

This alert is currently not used by any entitlement processes

Flows Using This Email Alert

Flow Name	Version	Description	Object	Active
Case Resolution Email Flow	1, 2, 4	Sends email to customer when Case status is changed to Closed	Case	✓

[Back To Top](#) [Always show me more records per related list](#)

Figure 10: The Case Resolution Email Alert configured with its linked template and recipient.

Testing: An existing case with a non-Closed status was updated to Closed. The flow fired correctly and the Case Resolution Email was delivered to the associated Contact's email address, confirming end-to-end automation.

7. Agentforce AI Implementation

The AI capability at the heart of this project is delivered through Agentforce, Salesforce's native AI agent platform, combined with Prompt Builder for structured, field-aware prompt generation. Two AI features were implemented: automatic case classification support and AI-suggested replies for agents.

7.1 How Agentforce Works in This Project

- A Case is created with a Subject, Description, Issue Type, and Priority.
- The Agentforce Agent, when triggered on the Case record, reads the case description via a Prompt Template.
- The Prompt Template asks the model to act as a customer support agent and draft a polite, professional, and helpful reply based on the Issue Type, Priority, and Description.
- The generated reply is displayed to the agent, either directly on the Case record (via the AI Suggested Response field) or through the conversational Agentforce Agent panel.
- The agent reviews, edits if needed, and sends the final reply to the customer.

7.2 Enabling Agentforce and Prompt Builder

- Setup → Einstein Setup → toggled Turn on Einstein to enable AI features org-wide.
- Setup → Prompt Builder → enabled Prompt Templates.

7.3 Prompt Template: AI Suggested Response

A Field Generation prompt template was created against the Case object, targeting a new long-text field, AI_Suggested_Response__c, added specifically to store the AI's draft reply.

Prompt used:

"You are a customer support agent. Based on the following case details, generate a polite, professional, and helpful reply. Issue Type: {!Case.Issue_Type__c}. Priority: {!Case.Priority}. Customer Description: {!Case.Description}. Provide a clear response that helps the customer resolve the issue."

The template was saved, activated, and assigned as the org default, then verified directly on the Case record by triggering the field's AI generation icon and confirming a coherent, context-aware reply was produced.

7.4 Agentforce Agent Configuration

The default Agentforce Agent was opened in Agent Builder and extended with a custom Topic and Action tailored to this use case:

Setting	Value
Topic Name	CST Topic
Classification Description	This topic is mainly about giving suggested responses for customer cases.
Scope	Providing solution/suggested responses for customers based on case details.

Setting	Value
Instructions	Run this topic when the user asks about a case's suggested reply; ask for the Case Id; then return the AI Suggested Response generated by the prompt template.

Agent Action

- Reference Action Type: Prompt Template; Reference Action: AI Suggested Response.
- Agent Action Label: AI Suggested Response.
- Instructions: first collect the Case Id from the user, then behave as a customer support agent and return a clear, helpful suggested reply.
- “Show in conversation” enabled so the generated reply displays directly in the chat with the agent.

The agent was activated and tested with prompts such as “Can you give a case solution response?”, which triggers the agent to request a Case Id, then returns the AI-generated suggested reply for that specific case — confirmed working end-to-end inside the AI-Powered CST System app.

7.5 Outcome

The combination of Prompt Builder and a custom Agentforce Topic/Action gives support agents a conversational, on-demand way to retrieve an AI-drafted reply for any case simply by supplying the Case Id — without leaving the Salesforce interface. Agent feedback on these suggestions is captured through the AI_Feedback__c object, closing the loop between AI output and measured usefulness.

8. Reporting & Monitoring

Two reports were built to give support managers real-time visibility into case load and agent performance, both accessible from the AI-Powered CST System app.

8.1 Open Cases by Priority

- Report Type: Cases; Filter: Status Not Equal To Closed.
- Columns: Case Number, Subject, Priority, Status, Case Owner.
- Grouped by Priority to give an immediate view of how many open cases fall into each priority tier.

8.2 Agent Performance Report

- Report Type: Cases; Filter: Status Equals Closed.
- Columns: Case Number, Owner, Priority, Created Date, Closed Date, Resolution Time (Hours).
- Grouped by Owner (agent), with Resolution Time summarized as an Average to reveal individual agent efficiency.

Report: Cases
Open Cases by Priority

Total Records: 5

Priority	Case Owner	Subject	Case Number	Status
High (2)	Bhimavarapu Lakshmi mahathi	Test Email Flow	00001027	Escalated
	Bhimavarapu Lakshmi mahathi	Test Validation	00001028	New
Subtotal				
Low (3)	OrgFarm EPIC	Design issue with mechanical rotor	00001024	New
	OrgFarm EPIC	Maintenance guidelines for generator unclear	00001016	New
	OrgFarm EPIC	Seeking guidance on electrical wiring installation for GC5060	00001002	New
Subtotal				
Total (5)				

Row Counts: ☐ Detail Rows: ☒ Subtotals: ☒ Grand Total: ☒

Figure 11: Open Cases by Priority and Agent Performance reports listed within the CST Reports folder.

8.3 Dashboard

A dashboard combining both reports was created to give managers a single-glance view of open case load by priority alongside agent resolution performance, refreshed in real time as case data changes.

Developer Edition

Search...

AI Powered CST System Home Cases AI Feedbacks SLA Policies Support Categories Reports Dashboards Contacts Accounts

Dashboards

All Dashboards

2 items

Search all dashboards...

New Dashboard New Folder

DASHBOARD NAME	DESCRIPTION	FOLDER	CREATED BY	CREATED ON	SUBSCRIBED
Enablement Dashboard	View data on how Enablement helps drive your business outcomes. This is your main dashboard for all Enablement analytics. Don't delete it. If you want to make changes to this dashboard, duplicate it.	Enablement Dashboard Spring '24	Automated Process	6/24/2026, 4:01 AM	
Enablement Dashboard	View data on how Enablement helps drive your business outcomes. This is your main dashboard for all Enablement analytics. Don't delete it. If you want to make changes to this dashboard, duplicate it.	Enablement Dashboard Summer '24	Automated Process	6/24/2026, 4:01 AM	

DASHBOARDS

Recent

Created by Me

Private Dashboards

All Dashboards

FOLDERS

All Folders

Created by Me

Shared with Me

FAVORITES

All Favorites

Figure 12: Dashboard visualizing open case distribution by priority and agent performance metrics.

9. Security & Access Control

Role-based access control was implemented using cloned profiles for the two primary user roles — Support Agent and Support Manager — alongside dedicated permission sets that grant additional, feature-specific access. The System Administrator profile was used as-is for full administrative access.

9.1 Support Agent Profile

Cloned from Standard User. Configured with:

- Case: Read, Create, and Edit enabled; Delete disabled.
- AI_Feedback__c: Create and Read enabled; Edit and Delete disabled.
- Account and Contact: Read-only access.
- Field-level security: Status and Priority editable; SLA Due Date and AI Suggested Response read-only.
- App access: AI Powered CST System enabled.

The screenshot shows the Salesforce Setup interface for the 'Support Agent' profile. The 'Profile Detail' section shows the profile name 'Support Agent', user license 'Salesforce', and creation/modification details. The 'Page Layouts' section contains a table with the following data:

Page Layouts	Standard Object Layouts	Location Group Assignment	Location Group Assignment Layout
Global	Global Layout (View Assignment)	Macro	Macro Layout (View Assignment)
Email Application	Not Assigned (View Assignment)	Object Milestone	Object Milestone Layout (View Assignment)
Home Page Layout	Home Page Default (View Assignment)	Operating Hours	Operating Hours Layout (View Assignment)
Account	Account Layout (View Assignment)	Opportunity	Opportunity Layout (View Assignment)
Alternative Payment Method	Alternative Payment Method Layout (View Assignment)	Opportunity Product	Opportunity Product Layout (View Assignment)
Appointment Invitation	Appointment Invitation Layout (View Assignment)	Order	Order Layout (View Assignment)
Asset	Asset Layout (View Assignment)	Order Product	Order Product Layout (View Assignment)
Asset Action	Asset Action Layout (View Assignment)	Payment	Payment Layout (View Assignment)
Asset Action Source	Asset Action Source Layout (View Assignment)	Payment Authorization	Payment Authorization Layout (View Assignment)
Asset Relationship	Asset Relationship Layout (View Assignment)	Payment Authorization Adjustment	Payment Authorization Adjustment Layout (View Assignment)
Asset State Period	Asset State Period Layout (View Assignment)	Payment Gateway	Payment Gateway Layout (View Assignment)
Assigned Resource	Assigned Resource Layout (View Assignment)	Payment Gateway Log	Payment Gateway Log Layout (View Assignment)
Associated Location	Associated Location Layout (View Assignment)		

Figure 13: Support Agent profile configuration showing object and field-level permissions.

9.2 Support Manager Profile

Cloned from Standard User. Configured with:

- Case: Create, Read, Edit, and Delete all enabled.
- Reports & Dashboards: Create and Subscribe access enabled.
- Field-level security: Priority, Owner, and SLA Due Date editable; AI Suggested Response read-only.
- App access: AI Powered CST System enabled.

Support Manager

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges \(0\)](#) |
 [Enabled Apex Class Access \(0\)](#) |
 [Enabled Visualforce Page Access \(0\)](#) |
 [Enabled External Data Source Access \(0\)](#) |
 [Enabled Named Credential Access \(0\)](#) |
 [Enabled External Credential Principal Access \(0\)](#) |
 [Enabled Custom Metadata Type Access \(0\)](#) |
 [Enabled Custom Settings Definitions Access \(0\)](#) |
 [Enabled Flow Access \(0\)](#) |
 [Enabled Agent Access \(0\)](#) |
 [Enabled Service Presence Status Access \(0\)](#) |
 [Enabled Custom Permissions \(0\)](#)

Profile Detail

Name: Support Manager
 User License: Salesforce
 Description: Not Assigned
 Created By: [Bhimavarapu Lakshmi mahathi](#), 7/7/2026, 10:05 AM
 Modified By: [Bhimavarapu Lakshmi mahathi](#), 7/7/2026, 10:16 AM

Page Layouts

Standard Object Layouts	Global	Location Group Assignment
Email Application	Global Layout (View Assignment)	Location Group Assignment Layout (View Assignment)
Home Page Layout	Home Page Default (View Assignment)	Macro Layout (View Assignment)
Account	Account Layout (View Assignment)	Object Milestone (View Assignment)
Alternative Payment Method	Alternative Payment Method Layout (View Assignment)	Operating Hours (View Assignment)
Appointment Invitation	Appointment Invitation Layout (View Assignment)	Opportunity (View Assignment)
Asset	Asset Layout (View Assignment)	Opportunity Product (View Assignment)
Asset Action	Asset Action Layout (View Assignment)	Order (View Assignment)
Asset Action Source	Asset Action Source Layout (View Assignment)	Order Product (View Assignment)
Asset Relationship	Asset Relationship Layout (View Assignment)	Payment (View Assignment)
Asset State Period	Asset State Period Layout (View Assignment)	Payment Authorization (View Assignment)
Assigned Resource	Assigned Resource Layout (View Assignment)	Payment Authorization Adjustment (View Assignment)
Associated Location	Associated Location Layout (View Assignment)	Payment Gateway (View Assignment)
Async Operation Log	Async Operation Log Layout	Payment Gateway Log (View Assignment)
		Payment Group (View Assignment)

Figure 14: Support Manager profile configuration with elevated Case and reporting permissions.

9.3 Permission Sets

Two permission sets extend the base profiles with feature-specific access, following the recommended Salesforce pattern of using profiles for baseline access and permission sets for additional capabilities.

Permission Set	Grants	Assigned To
AI Feature Access	Edit AI-related Case fields; Create/Read on AI_Feedback__c; Access Einstein Features; Use Agent Assist	Support Agents
Report Access	Create and Customize Reports; Subscribe to Reports	Support Managers

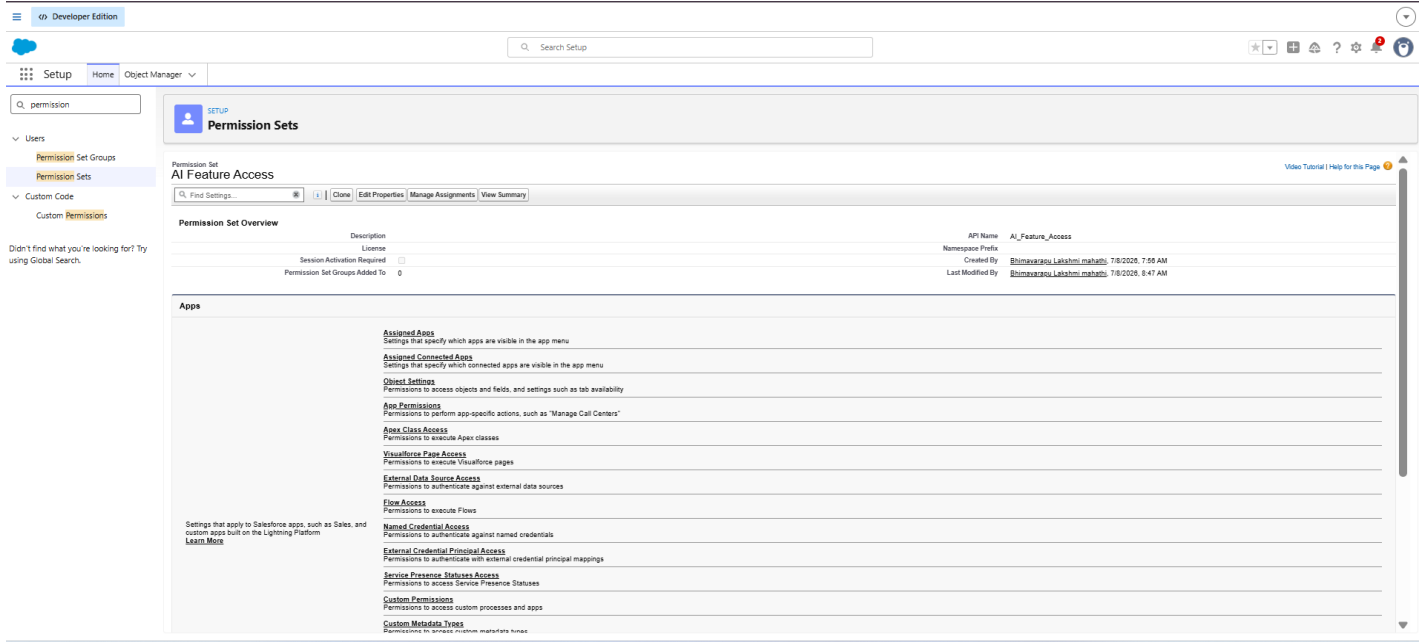


Figure 15: The AI Feature Access and Report Access permission sets configured in Setup.

9.4 Role – Profile – Permission Set Summary

Role	Profile	Permission Set
Support Agent	Support Agent	AI Feature Access
Support Manager	Support Manager	Report Access
Admin	System Administrator	All (native)

9.5 Users

Three users were created to represent the distinct roles in the system, each assigned the appropriate profile and permission set:

User	Profile	Permission Set Assigned
Admin User	System Administrator	—
Support Agent	Support Agent	AI Feature Access
Support Manager	Support Manager	Report Access

Developer Edition

SetupHomeObject Manager

Q Users

Users

Analytics Groups
Permission Set Groups
Permission Sets
Profiles
Public Groups
Queues
Roles

User Management Settings

Users

Feature Settings

Data.comProspectorUsers

On this page you can create, view, and manage users.
To get more licenses, use the Your Account app. Let's Go

All Users

New UserReset Password(s)Add Multiple Users

Action	Full Name ↑	Alias	Username	Role	Active	Profile
[Edit]	Ajeet Auroor	ajeet	ajeeest.mahath@202@gmail.com		✓	Succot Asset
[Edit]	Chatter Expert	Chatter	chattv.DDd00000Yasbvrai.cDuDqk3m@d.chatter.salesforce.com		✓	Chatter Free User
[Edit]	EinsteinServiceAgentUser	einstein	pase.sureent_ajant@DDd000000YasbvraIDBZ4TJ80.ausf		✓	Einstein Agent User
[Edit]	EPIC_CpaFarm	OEPIC	poic.B1c7111Idaf@cpafarm.salesforce.com		✓	System Administrator
[Edit]	Lakshmi mahathi_Bhimavaraju	lks	lakshmimahatibhimavaraja.ad@3o3e1G61b@eventforce.com		✓	System Administrator
[Edit]	User_Admin	admin	admn.est.mahath@202@gmail.com		✓	System Administrator
[Edit]	User_Integration	inteq	integration@DDd000000Yasbvrai.com		✓	Analytix Cloud Integration User
[Edit]	User_Security	sbs	anishbassecurity@DDd000000Yasbvrai.com		✓	Analytix Cloud Security User

New UserReset Password(s)Add Multiple Users

Figure 16: All three users listed in Setup → Users, with profiles assigned.

Setup

Users

User Profile Help for this Page

Salesforce now sends emails from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

Permission Set Assignments | Permission Set Assignments Activation Required | Permission Set Group Assignments | Permission Set License Assignments | Personal Groups | Public Group Memberships | Queue Memberships | Teams | Managers in the Role Hierarchy | OAuth Apps | Third-Party Account Links | Built-in Authentication | Installed Mobile Apps | Authentication Settings for External Systems | Login History | User Provisioning Activity

User Detail

User		Action	Role
Name	Admin User	[Edit] [Sharing] [Reset Password] [Freeze] [View Summary]	User License Salesforce
Alias	admin		Profile System Administrator
Email	admin.cs@demo.com [Verify]		Active ✓
Username	admin.csst.mahath@202@gmail.com		Marketing User ☐
Nickname	User1783523832034902080 [x]		Offline User ☐
Title			Knowledge User ☐
Company			Flow User ☐
Department			Service Cloud User ☐
Division			Site.com Contributor User ☐
Address			Site.com Publisher User ☐
Time Zone	(GMT-07:00) Pacific Daylight Time (America/Los_Angeles)		WDC User ☐
Locale	English (United States)		Mobile Push Registrations View [x]
Language	English		Data.com User Type [x]
Delegated Approver			Accessibility Mode (Classic Only) ☐ [x]
Manager			Debug Mode ☐ [x]
Receive Approval Request Emails	Only if I am an approver		High-Contrast Palette on Charts ☐ [x]
Federation ID			Load Lightning Pages While Scrolling ✓ [x]
App Registration: One-Time Password Authenticator [x]			Send Apex Warning Emails ☐
App Registration: Salesforce Authenticator [x]			Salesforce CRM Content User ✓
Security Key (U2F or WebAuthn) [x]			Receive Salesforce CRM Content Email Alerts ✓
Lightning Login [x]			Receive Salesforce CRM Content Alerts as Daily Digest ✓
Temporary Verification Code (Expires in 1 to 24 Hours) [Generate] [x]			Make Setup My Default Landing Page ☐
			Quick Access Menu ✓
			Development Mode ☐
			Show View State in Development Mode ☐

Figure 17: Admin User record configured with the System Administrator profile.

The screenshot shows the Salesforce Setup page for the 'Support Agent' user. The user is configured with the 'Support Agent' profile and is marked as 'Active'. The user's email is 'agent.cst@demo.com' and their username is 'agent.cst.mahesh2020@gmail.com'. The user is assigned the 'Support Agent' role, which includes permissions for Salesforce, Marketing User, Knowledge User, Flow User, Service Cloud User, Site.com Contributor User, Site.com Publisher User, WEC User, Mobile Push Registrations, Data.com User Type, Accessibility Mode (Classic Only), Debug Mode, High-Contrast Palette on Charts, Load Lightning Pages While Scrolling, Salesforce CRM Content User, Receive Salesforce CRM Content Email Alerts, Receive Salesforce CRM Content Alerts as Daily Digest, Make Setup My Default Landing Page, Allow Forecasting, No MRU Updates, Call Center, Phone, and Extension.

User Detail

Field	Value	Field	Value
Name	Support Agent	Role	Salesforce
Alias	agent	User License	Support Agent
Email	agent.cst@demo.com	Profile	Support Agent
Username	agent.cst.mahesh2020@gmail.com	Active	☒
Nickname	User17835240810227570110	Marketing User	☐
Title		Offline User	☐
Company		Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☐
Address		Site.com Contributor User	☐
Time Zone	(GMT-07:00) Pacific Daylight Time (America/Los_Angeles)	Site.com Publisher User	☐
Locale	English (United States)	WEC User	☐
Language	English	Mobile Push Registrations	☒
Delegated Approver		Data.com User Type	☐
Manager		Accessibility Mode (Classic Only)	☐
Receive Approval Request Emails	Only if I am an approver	Debug Mode	☐
Federation ID		High-Contrast Palette on Charts	☐
App Registration: One-Time Password Authenticator	☐	Load Lightning Pages While Scrolling	☒
App Registration: Salesforce Authenticator	☐	Salesforce CRM Content User	☒
Security Key (U2F or WebAuthn)	☐	Receive Salesforce CRM Content Email Alerts	☒
Lightning Login	☐	Receive Salesforce CRM Content Alerts as Daily Digest	☒
Temporary Verification Code (Expires in 1 to 24 Hours)	☐	Make Setup My Default Landing Page	☐
		Allow Forecasting	☐
		No MRU Updates	☐
		Call Center	☐
		Phone	☐
		Extension	☐
		Fax	☐

Figure 18: Support Agent user record configured with the Support Agent profile.

The screenshot shows the Salesforce Setup page for the 'Support Agent' user, displaying the 'Permission Set Assignments' section. The user is assigned the 'AI Feature Access' permission set, which is active and expires on 7/8/2025. The user is also assigned the 'Support Agent' profile and is marked as 'Active'.

Permission Set Assignments

Action	Permission Set Name	Date Assigned	Expires On
Del	AI Feature Access	7/8/2025	

Permission Set Assignments: Activation Required

No records to display

Permission Set Group Assignments

No records to display

Permission Set License Assignments

No records to display

Personal Groups

No personal groups specified

Public Group Membership

No records to display

Queue Membership

No records to display

Team

No records to display

Managers in the Role Hierarchy

No records to display

OAuth Apps

No approved access tokens

Figure 19: AI Feature Access permission set assignment on the Support Agent user.

Final verification confirmed that each user has exactly one profile, is marked Active, has a unique username, and that the assigned profile correctly matches the intended role — satisfying the security design of the system.

10. Functional & Performance Testing

Testing was performed throughout the development phase to verify that each automation, validation rule, and AI feature behaved as designed. The following scenarios were executed and verified directly in the Salesforce Developer Org:

- Case creation across Low, Medium, and High priority scenarios, confirming correct automatic classification and SLA due date calculation.
- Validation rule testing: attempting to save a High priority case with a blank SLA Due Date was correctly blocked; supplying a date allowed the save to succeed.
- Agentforce testing: triggering the AI Suggested Response action with a valid Case Id returned a coherent, context-appropriate reply drafted from the case's Issue Type, Priority, and Description.
- AI Feedback testing: logging Helpful / Not Helpful feedback against a case correctly created an AI_Feedback__c record linked to that case.
- Email alert testing: updating a case's Status to Closed triggered the Case Resolution Email flow, and the email was delivered to the Contact's email address.
- Reports and dashboard testing: confirmed that Open Cases by Priority and Agent Performance reports reflected live case data accurately, and that the dashboard refreshed correctly.
- Security testing: logged in as each of the three test users to confirm object, field, and app-level access matched the intended profile and permission set configuration.

Performance testing confirmed that record-triggered flows executed instantly on save, and that reports and dashboards rendered within a few seconds even as case volume increased, with no noticeable delay for end users.

11. Maintenance, Monitoring & Troubleshooting

The Salesforce AI-Powered Customer Support Ticketing System requires ongoing maintenance and monitoring to ensure smooth operation, data accuracy, and reliable automation. Regular maintenance activities include reviewing and updating custom objects, fields, picklist values, and flows to align with changing business requirements. Profiles and permission sets are periodically audited to ensure proper access control and compliance with security best practices.

System monitoring focuses on tracking case volumes, SLA compliance, and automation performance using reports and dashboards. Flow executions are reviewed to confirm that case classification, priority assignment, and SLA calculations are functioning as expected. AI feedback data is monitored to evaluate the effectiveness of system-generated suggestions and identify areas for improvement.

Troubleshooting involves analyzing flow errors, validation failures, and data inconsistencies when issues occur. Administrators review debug logs, flow error emails, and record history to identify root causes. Corrective actions may include updating flow logic, adjusting field-level security, refining decision conditions, or correcting data configurations. This structured approach ensures system stability, minimizes downtime, and supports continuous improvement of the support process.

12. Advantages & Disadvantages

12.1 Advantages

- Centralized management of all customer support cases in a single Salesforce app.
- Reduced manual effort through automated case classification, priority assignment, and SLA calculation.
- Faster, more consistent agent responses through AI-suggested replies via Agentforce.
- Improved SLA compliance visibility for support managers through real-time reports and dashboards.
- Strong data protection through role-based profiles, permission sets, and field-level security.
- A built-in feedback loop that measures and can improve AI suggestion quality over time.
- Built entirely on native Salesforce Admin and Flow features, keeping the solution low-cost and easy to maintain.

12.2 Disadvantages

- Requires an active Salesforce license and Agentforce/Einstein feature enablement, which may carry cost implications in a production org.
- Fully dependent on internet connectivity as a cloud-based platform.
- AI-suggested replies still require agent review before sending, so full automation of responses is intentionally not supported.
- Limited offline access, consistent with other cloud-native Salesforce applications.

13. Conclusion

The Salesforce AI-Powered Customer Support Ticketing System successfully demonstrates how intelligent automation and structured data modeling can improve the efficiency of customer support operations. By leveraging standard and custom Salesforce objects, record-triggered flows for case classification, priority assignment, and SLA tracking, and a well-defined security model, the system reduces manual effort and ensures timely issue resolution.

The inclusion of Agentforce-powered agent assistance and a structured feedback mechanism enhances agent productivity while enabling continuous improvement of the system's AI suggestions. Overall, the project delivers a scalable, maintainable, and real-world support solution, and provides strong hands-on experience with core Salesforce Admin, Flow automation, and Agentforce concepts — making it well suited for both practical implementation and interview demonstration.

14. Future Scope

- Extend Agentforce with multi-turn conversational case triage, allowing customers to interact directly with the agent before a human takes over.
- Introduce a knowledge base integration so AI-suggested replies can cite and link relevant help articles.
- Add escalation automation that reassigns or flags cases automatically when an SLA is close to breaching.
- Build a mobile-optimized agent experience for on-the-go case handling.

- Expand the AI feedback loop into a lightweight analytics dashboard tracking suggestion accuracy trends over time.
- Integrate omni-channel case creation (email-to-case, chat, and social) to broaden how customers can raise issues.

15. Appendix

Item	Details
Source	Salesforce Configuration — Objects, Flows, Validation Rules, Agentforce, Prompt Builder
Dataset	Sample Case, Contact, and Account records created during testing
Tools Used	Salesforce Developer Org, Flow Builder, Agentforce Agent Builder, Prompt Builder, Reports & Dashboards
Team	Bhimavarapu Lakshmi Mahathi, Kudumula Avinash Reddy, MANDA Ijak Babu, Thiramdasu Sridevi, Sravanthi Uppalapati

Thank You